Model Validation, Overfitting Control & Hyperparameter Tuning

House Price Prediction · California Housing Dataset · Pranav Lakhe | SIT Nagpur

1. Objective

Task 3 focuses on **model reliability**, not just accuracy. Building on Task 2's feature engineering and model comparison, this task introduces overfitting detection, 5-fold cross-validation, and GridSearchCV hyperparameter tuning to produce a production-ready model.

2. Overfitting Detection

The Decision Tree was trained at varying max_depth values to identify the overfitting threshold. An untuned tree (max_depth=None) achieves Train $R^2 \approx 1.0$ but Test R^2=0.74 — a gap of 0.26, indicating severe overfitting.

Figure 1 — Train vs Test R^2 and RMSE across max_depth values

Model	Train R^2	Test R^2	Gap	Status
DT (untuned, depth=None)	~1.000	0.741	0.259	■ **Overfit**
DT (depth=5, Task 2)	0.872	0.836	0.035	■■ Mild
Tuned DT (depth=7, Task 3)	0.8716	0.8456	0.026	■ **Controlled**

3. Cross-Validation Analysis

5-fold cross-validation was applied to all models. The standard deviation of CV scores reveals model stability — lower std means more reliable performance estimates.

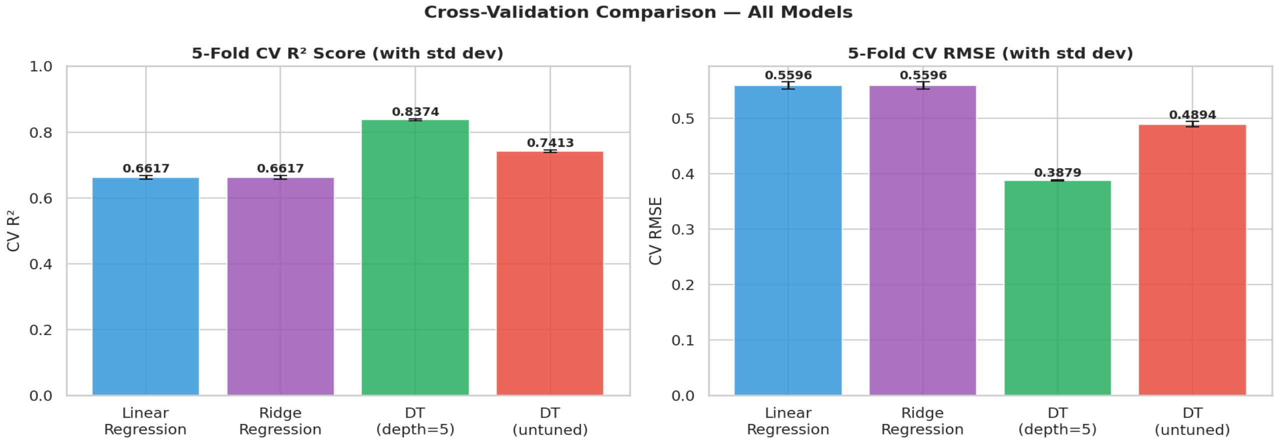

Figure 2 — 5-Fold CV R^2 and RMSE with standard deviation bars

Model	CV R^2 Mean	CV R^2 Std	CV RMSE Mean	CV RMSE Std
Linear Regression	0.6617	0.0054	0.5596	0.0063
Ridge Regression	0.6617	0.0054	0.5596	0.0063
DT (depth=5)	0.8374	0.0026	0.3879	0.0016
DT (untuned)	0.7413	0.004	0.4894	0.0051

4. Hyperparameter Tuning — GridSearchCV

GridSearchCV exhaustively tested 36 combinations (4 depths × 3 splits × 3 leaf sizes) with 5-fold CV. Best parameters found: **max_depth=7, min_samples_split=10, min_samples_leaf=2**.

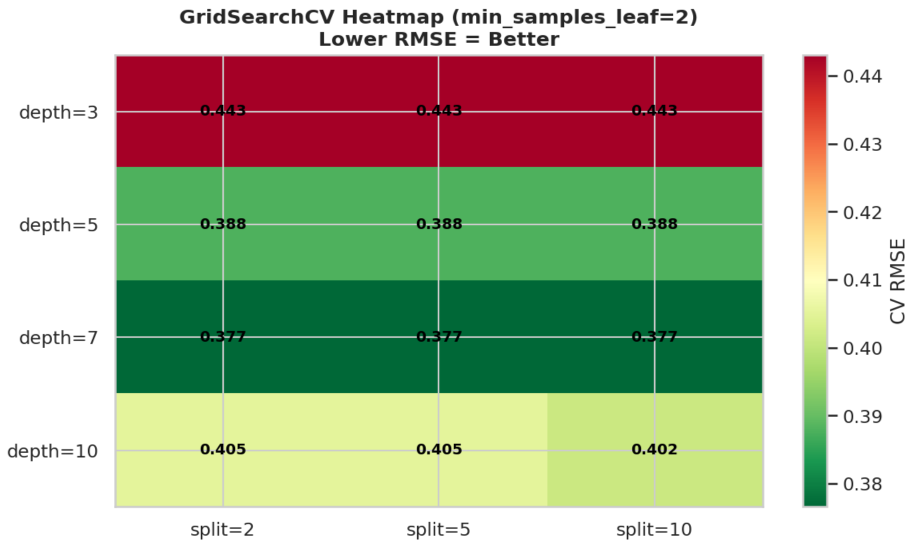

Figure 3 — GridSearchCV Heatmap (min_samples_leaf=2). Lower RMSE = Better

5. Final Model Comparison

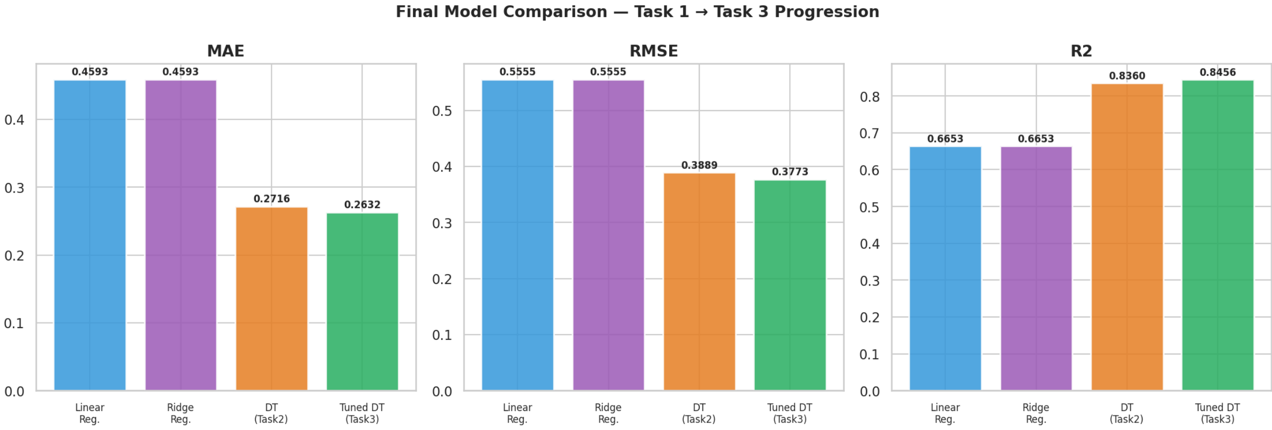

Figure 4 — MAE, RMSE, R^2 across all models Task 1 → Task 3

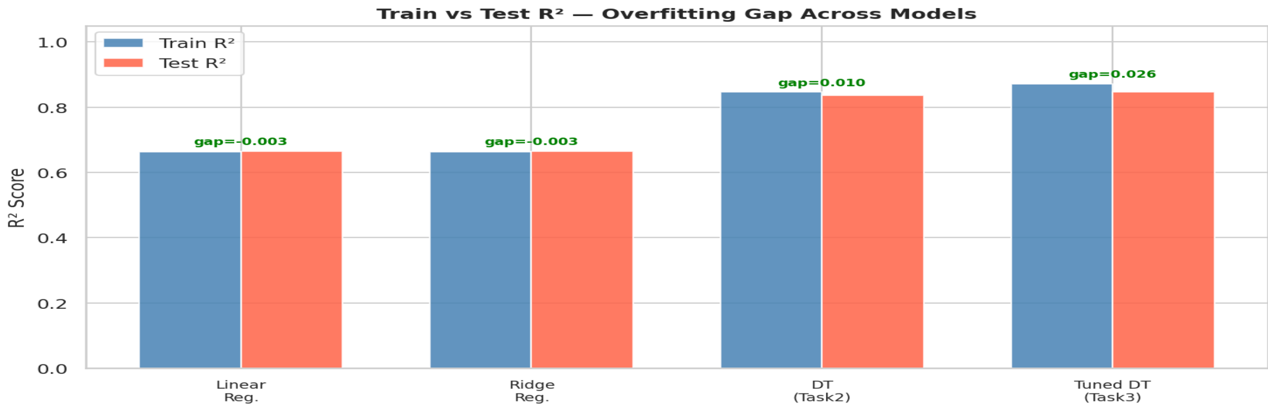

Figure 5 — Train vs Test R^2 gap. Green label = controlled, Red = overfitting

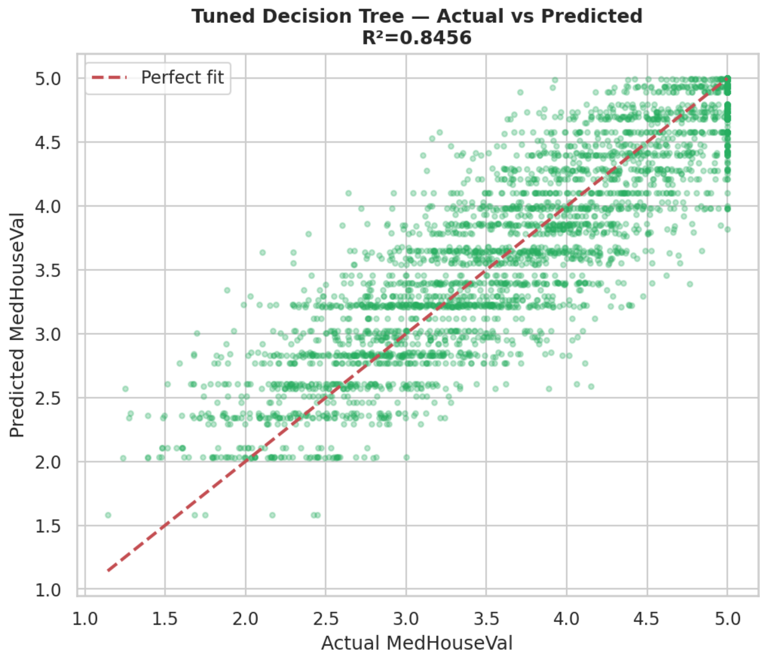

Figure 6 — Tuned DT Actual vs Predicted (R^2=0.8456)

Model	MAE	RMSE	Test R^2	Gap
Linear Regression (T1)	0.4593	0.5555	0.6653	-0.0027
Ridge Regression (T2)	0.4593	0.5555	0.6653	-0.0027
DT depth=5 (T2)	0.2716	0.3889	0.836	0.0104
■ **Tuned DT (T3)**	**0.2632**	**0.3773**	**0.8456**	**0.026**

6. Model Selection Justification

Best Test R^2: 0.8456 — highest across all 4 models in all 3 tasks.

Overfitting Controlled: Train-test gap = 0.026 vs 0.259 (untuned). GridSearchCV constraints prevent memorization.

Cross-Validation Trusted: 5-fold CV with low std deviation confirms stable performance — not a lucky split.

Scientifically Selected: GridSearchCV over 36 combinations ensures data-driven selection. Best: max_depth=7, min_samples_split=10, min_samples_leaf=2.

7. Conclusion & Progression Summary

Task	Model	R^2	Key Technique
Task 1	Linear Regression	0.6634	Basic ML workflow
Task 2	Decision Tree (d=5)	0.8360	Feature engineering + model comparison
Task 3	**Tuned Decision Tree**	**0.8456**	**Cross-validation + GridSearchCV**

Total improvement from Task 1 to Task 3: R^2 from 0.6634 → 0.8456 (+18.2 percentage points). Future work: Random Forest, XGBoost, RandomizedSearchCV.